



Phenakite

△ **Large, finely-formed** gem quality phenakite crystals on matrix

Phenakite's name originates from the Greek for “deceiver” – see box, below. It can be colourless, but more often is translucent grey or yellow and, occasionally, pale rose-red. Phenakite is found in high-temperature pegmatites and in mica-schists, often accompanied by quartz, chrysoberyl, apatite, and topaz. Its crystals are mainly rhombohedrons, and sometimes short prisms. Transparent crystals are faceted for collectors. Its indices of refraction are higher than topaz, and its brilliance approaches that of diamond.

Specification

Chemical name Beryllium silicate | **Formula** Be_2SiO_4 | **Colours** Colourless, white | **Structure** Hexagonal/trigonal | **Hardness** 7.5–8 | **SG** 3 | **RI** 1.65–1.67 | **Lustre** Vitreous | **Streak** Colourless | **Locations** Russia, Norway, France, USA



Large crystal | Rough | This large single crystal of phenakite with adhering matrix at the base shows perfect phenakite crystal form.



Brazilian phenakite | Cut | Cut in a fancy cushion with multiple, layered facets, this stunning Brazilian stone weighs 29.80 carats.

Phenakite and quartz

The great deceiver

Phenakite has a well-deserved reputation for deception, also the basis for its name – it is difficult to distinguish it from colourless quartz, both in its appearance and its technical specifications. Mineralogists can use various solutions to tell the two substances apart, including testing their specific gravity (quartz's is slightly lower, at 2.65, while phenakite's is 3) or their hardness (phenakite is slightly harder). The latter can be tested by conducting a scratch test on a piece of quartz.



Clear quartz crystals This specimen of rough rock crystal could potentially be mistaken for phenakite.



Burmese phenakite | Cut | This extremely fine, totally colourless 25.57-carat brilliant oval-cut phenakite from Myanmar is nearly an inch in length.

Table facet